

Ashley Heckman

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School Address
Ithaca, NY 14850

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Winter Garden, FL 34787

Education

Cornell University (Ithaca, NY) M.Eng. Electrical and Computer Engineering	Jan 2024 – Dec 2024
Cornell University (Ithaca, NY) - B.S. Electrical and Computer Engineering - Computer Science Minor - Dean's List	August 2020 – May 2024
Brighton High School (Brighton, MI) - High School Diploma - Top 2%	August 2016 – May 2020

Selected Projects

Plant Watering System

Using a Raspberry Pi and external sensors, autonomously water the selected plant. Includes a nice user interface for the user to select their plant and watering mode.

Computer Processor

Develop a single-core and multi-core pipelined processor with bypassing capable of running basic programs.

Autonomous Robot

Using an Arduino and external sensors, navigate through a maze until two treasures are found and analyzed.

Mandelbrot Set Acceleration

Accelerate the computation of the Mandelbrot set on an FPGA using Vivado HLS and optimization techniques.

Ecosystem Simulator

Using the DE1-SoC FPGA, display a marine ecosystem that models the behaviors of various sea creatures.

Skills

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|-----------|---------------------------------------|----------------------------|
| • C | • Altera Quartus | • Waveform Debugging |
| • C++ | • Xilinx Vivado | • Github |
| • Linux | • Microcontroller Programming | • Microsoft Office |
| • Python | • Computer Architecture | • Communication & Teamwork |
| • Verilog | • Oscilloscopes & Function Generators | • Problem Solving |

Work Experience

Systems Engineer Intern D - Created a document detailing what is required from the satellite operation center - Reviewed hardware requirements - Helped configure virtual machines - Created sequence diagrams to describe code - Found alternative parts for products with long lead times - Installed rack components & labelled and wired rack cables	L3Harris Technologies (Palm Bay, FL)	May 2023 – Dec 2023
Embedded Software Engineer - Design electronic systems to control a hyperloop pod - Build circuits using Arduino and sensors; write software to control the state of the pod; read sensor data; communicate data to GUI	Cornell Hyperloop (Ithaca, NY)	August 2022 – Jan 2024
Embedded Operating Systems TA - Cover a lab section. Guide students through the lab, answer questions, and debug any issues that arise - Grade homework, lab reports, and final projects	Cornell University (Ithaca, NY)	August 2023 – Dec 2024
Undergraduate Resident Fellow Serve as a peer mentor; help build a safe and welcoming community; plan/promote a variety of programs and activities; provide assistance, resources, and referral information to students	Keeton House (Ithaca, NY)	August 2022 – May 2024